# Haoyuan Zheng

hyzh@umich.edu | 734-210-4531 | Ann Arbor, MI 48105

## EDUCATION

**University of Michigan, Ann Arbor** Sep. 2026 – present

*B.S.E. in Computer Engineering*

**Shanghai Jiao Tong University (SJTU)** Sep. 2024 – Jun. 2028 (exp.)

*B.S.E. in Electrical and Computer Engineering | GPA 3.68/4.0*

Coursework: Data Structures and Algorithms, Honors Mathematics IV (A+), Intro to Circuits (A+), Physics II (A), Signals & Systems (A-), EECS 245, EECS 311, and EECS 370 (in progress)

Awards: SJTU Class C Excellence Scholarship (2025); Cheng Family Scholarship (2026)

## RESEARCH & PROJECT EXPERIENCE

**Visual SLAM and 3D Vision, IRMV, SJTU** Jul. 2026 – Aug. 2026

*Advisor: Prof. Hesheng Wang*

- Built an object-oriented visual SLAM system in Python: SIFT/ORB features, geometric matching, relative pose estimation, PnP tracking, triangulation, and sparse bundle adjustment.
- Ran controlled experiments on frame sampling, keyframe selection, feature density, and match filtering, comparing reprojection error, point-cloud density, and runtime.
- Extended the pipeline toward dense 3DGS-SLAM with Depth Anything V2 and ADD-SLAM; diagnosed failures from limited viewpoint coverage, depth-scale inconsistency, and camera-pose drift.

**Rock Pick Pose Estimation, MRover (Michigan Mars Rover Team)** 2026 – present

- Developing keypoint-based pose estimation of a rock pick for autonomous retrieval: a downward-facing USB camera provides images, and the model outputs position and keypoints for the grasping subsystem.
- Built a ~10,000-frame dataset, using CoTracker for auto-labeling, and trained a YOLO-based keypoint model on it.
- Integrating with the rover software stack through ROS topics and the TF tree.

**C2-MANS: Neuro-Symbolic Multi-Agent Novel Generation, SJTU** Nov. 2025 – Apr. 2026

*Advisor: Prof. Jiong Lou*

- Designed a CrewAI pipeline of four specialized agents with Pydantic state models to keep long-form narratives logically consistent across chapters.
- Implemented structured state layers (RelationshipGraph, TimeLedger) with persistent storage for autonomous generation of 10-chapter novellas.

## SKILLS

**Programming:** Python, C/C++, MATLAB

**Vision & Robotics:** Visual SLAM, multi-view geometry, 3D reconstruction, object detection and keypoint models (YOLO), ROS, SolidWorks, prototyping

**Tools:** PyTorch, Pandas, Git, Linux, LaTeX